

Response to Reviewers

Manuscript: “Performance Analysis of Floating Solar Photovoltaic Systems for Humid and Saline Conditions: A Coupled Thermal, Soiling and Degradation Assessment”

Revised title: “Lifetime Performance and Degradation of Floating Photovoltaic Systems Under Humid and Saline Conditions”

I thank the reviewer for a careful and constructive assessment. The revision addresses every comment. The largest changes are:

1. The IEC 61701/PID terminology is corrected throughout, and the mitigation factors are presented as explicit assumptions (new Table IV) and sampled in the uncertainty analysis.
2. A 1,000-sample Monte Carlo over 26 uncertain parameters is added (new Section III-I, Section IV-H, Table X, Fig. 12). It is reported separately from the 20-year weather ensemble.
3. New threshold analyses (salt, chloride and dew-point rise; Fig. 9), a mitigation comparison (Table VIII), economic break-even values (Section IV-I), a dew-loss sensitivity (Section IV-C), a literature comparison table (Table I) and a weather-generator check (Table III) are added.
4. Results are reframed as model-based scenario predictions throughout, and the scope is restricted explicitly to module-level electrical degradation.

One important outcome of the new uncertainty analysis is that the freshwater FPV lifetime advantage is *not* robust (positive in 73% of samples), whereas the saline lifetime deficit is (negative in all but 4% of samples). The abstract, results and conclusion now state this. All numbers in the manuscript and in this letter are generated by one script from the open-source model, so they are mutually consistent.

A correction to the generator also changed the baseline numbers slightly. The weather check requested by the reviewer revealed that the synthetic year over-estimated mean wind speed by about 0.5 m/s and mean RH by about 2.6 points relative to its input normals. Both biases have been removed (wind variability is now mean-preserving, and the daily dew point is adjusted per month). The baseline saline lifetime deficit therefore changes from -4.9% to -4.3% , and the saline degradation rate from 1.24 to 1.16%/yr. None of the qualitative conclusions changes.

Section 3: IEC 61701 associated with PID resistance

Comment 1 (terminology). *IEC 61701 is the salt-mist corrosion standard, not a PID qualification; IEC TS 62804-1 is the PID standard. Correct the terminology throughout.*

Response. Agreed; this was an error. The mitigated case is now described as “PID-resistant module technology (qualified with IEC TS 62804-1:2025) combined with salt-mist hardening (qualified with IEC 61701:2020)”. The two effects are kept separate in the model, the text, the figures and the code. The reference to IEC TS 62804-1 is updated to the 2025 edition.

Changes: Abstract; Sections I, III-G, IV-E, V-B; Table IV; captions of Fig. 7 and Tables VII–VIII; bibliography; code (`with_mitigation` now has separate `pid` and `salt_mist` switches).

Comment 2 (μ_{PID} and μ_{C}). *The reductions $\mu_{\text{PID}} = 0.9$ and $\mu_{\text{C}} = 0.5$ are not justified by certification. Add a table of standards and assumptions, or perform a sensitivity analysis.*

Response. Agreed. New Table IV lists each mitigation, its qualification standard, how it enters the model and the assumed value with its Monte Carlo range (μ_{PID} : 0.5–0.95; μ_{C} : 0.2–0.7). The text now states that qualification standards define test procedures and do not establish a numerical reduction of field degradation. With the factors sampled, the 95% interval of the mitigated saline lifetime difference to land is $[-7.4, 3.6]\%$. No quantitative literature value supports a specific reduction, so the factors are treated purely as assumptions.

Changes: Section III-G and Table IV (new); Section IV-H and Table X.

Section 4: Environmental model largely assumed

Comment 3 (wording). *Present the 1.24%/yr saline degradation as a scenario-model result under assumed environmental parameters.*

Response. Agreed. The results section now opens by stating that all results are model predictions under the baseline assumptions. Absolute statements are rephrased (“Under the baseline assumptions, total degradation is ...”; “the predicted degradation rate”). Table II is retitled “Baseline Scenario Parameters (Assumptions)”, and the Monte Carlo interval for the saline degradation rate is reported alongside the baseline value (Table X).

Changes: Abstract; Sections II, IV (opening sentence), IV-D, VI.

Section 5: Synthetic weather is not validation

Comment 4 (weather validation). *Validate against measured data or, at minimum, compare the synthetic weather statistics with measured data (GHI, temperature, RH, rainfall, wind; RMSE, bias, correlation).*

Response. Measured hourly data for the site could not be obtained for this revision. New Table III reports the MBE, RMSE and correlation of 20 generated years against the monthly normals the generator is built from. This check uncovered and led to the correction of the wind and RH biases described above. The manuscript states explicitly that this is a check of the generator against its inputs, not an external validation; that the RH match is exact by construction; and that GHI (through the calibrated clearness index) has not been checked against measurements. The model accepts measured hourly weather (**-weather**), and validation against measured meteorological and FPV data is named as the next step. I acknowledge that this comment is only partly addressed.

Changes: Section III-A and Table III (new); Section V-C (Limitations); generator correction in `fpv_analysis/climate.py`.

Section 6: Degradation model too empirical; module vs. system degradation

Comment 5 (scope). *Separate electrical/module degradation from FPV system degradation (structural fatigue, moorings, storm damage).*

Response. Agreed. A “Scope” paragraph at the end of Section I states that the degradation results describe the electrical degradation of the PV modules only, and that floating-system degradation is outside the model. Recent work on structural corrosion and mechanical degradation of FPV components is cited. The limitations list the missing physics (PID saturation and recovery, mechanism interactions, interconnect failure).

Changes: Section I (Scope); Section V-C; “module” added to degradation labels in Tables VII–VIII and Fig. 7.

Section 7: Dew model

Comment 6 (4% optical loss). *Justify the 4% dew optical loss or show a sensitivity for 0, 2, 4 and 6%.*

Response. The value is now labelled as an engineering assumption. New Section IV-C shows that varying it over 0–6% changes the year-1 yield difference to land by less than 0.05 percentage point (freshwater: 1.64–1.66%; saline: –1.10––1.06%), because condensate evaporates within the first hours after sunrise. The dew loss is also sampled (0–6%) in the Monte Carlo. Dew matters through time of wetness, not as an optical loss.

Changes: Section III-C; Section IV-C (new).

Section 8: Salt/soiling model

Comment 7 (salt deposition sensitivity). *Show salt deposition $\rightarrow$ first-year yield and salt deposition $\rightarrow$ lifetime energy.*

Response. Added as Fig. 9(a). Saline FPV out-yields land PV in year 1 only below about 35 mg/m²/day of salt deposition, and its 25-year energy stays below land at every salt level (-0.9% at zero salt). The salt deposition, soiling coefficients and cementation factor are also sampled in the Monte Carlo; the saline salt deposition is the third most influential parameter for the saline lifetime difference. Field evidence that sea-salt deposits dominate on coastal FPV modules [Ahmad *et al.*, 2026] is now cited.

Changes: Section IV-F and Fig. 9 (new); Section III-D.

Section 10: Table III caption

Comment 8 (non-additive losses). *State that the sequential loss percentages are not directly additive.*

Response. Done. The table caption now reads “Percentages Are Sequential Losses Relative to the Preceding Stage and Are Not Directly Additive”.

Changes: Table VI caption; Fig. 4 caption.

Section 11: Freshwater gain and parameter uncertainty

Comment 9 (weather vs. parameter uncertainty). *The $\pm 0.08\%$ from 20 weather realizations is stochastic weather uncertainty only; report parameter uncertainty for the absolute yield.*

Response. Agreed. Weather variability and parameter uncertainty are now treated and reported separately (Section IV-H). The parameter uncertainty is much larger. The year-1 yield difference of freshwater FPV has a 95% interval of [–1.8, 5.3]% (positive in 76% of samples), and its lifetime difference [–3.6, 6.4]% (73%). Table X also gives intervals for the absolute yields, degradation rates and LCOEs. The headline claim is softened accordingly: the freshwater advantage is a baseline result that depends mainly on the FPV heat-loss coefficients and on how dusty the land alternative is.

Changes: Abstract; Sections III-I, IV-H (new); Table X; Fig. 12; Sections V and VI.

Section 12: LCOE break-even

Comment 10 (break-even analysis). *Add break-even values for CAPEX, land cost, evaporation value and salt deposition.*

Response. Added (Section IV-I). Because LCOE is linear in CAPEX and in the water value, the break-even values follow in closed form:

- Parity with land PV requires a CAPEX of 34.3 INR/Wp for freshwater FPV and 29.1 INR/Wp for saline FPV.
- Land PV would need a further 10.5 or 23.6 INR/Wp of land-related cost to match them.
- Freshwater parity requires a water value of about 93 INR/m³.
- No salt level makes saline FPV competitive at the assumed CAPEX.

In the Monte Carlo, FPV is more expensive than land PV in 99.2% (freshwater) and 100.0% (saline) of samples.

Changes: Section III-H (equation (15) with water benefit); Section IV-I (new).

Section 13: Evaporation benefit

Comment 11 (monetize water). *Monetize the avoided evaporation and report the LCOE including the water benefit.*

Response. Done. The LCOE definition now includes an optional water benefit $B_y = c_w V_w$. At 10, 20 and 40 INR/m³, the freshwater LCOE falls from 3.54 to 3.47, 3.39 and 3.24 INR/kWh. The water values are illustrative and are given as a range rather than as a single tariff.

Changes: Equation (15); Section IV-I.

Section 14: Literature positioning

Comment 12 (comparison table). *Add a systematic comparison with recent work in a feature table.*

Response. Added as Table I, with 19 new references. The closest prior work is now discussed explicitly: Mannino *et al.* (2023) applied existing degradation models to offshore meteorological data, and Ajmal *et al.* (2025) added salinity to calibrated climatic degradation models for offshore modules. The novelty claim is narrowed to the coupling, in one microclimate, of the yield benefits with the humidity-, salt- and PID-driven penalties, and to the freshwater-versus-saline comparison with lifetime energy and cost. Recent FPV thermal measurements (Lindholm 2021; Dörenkämper 2024; Nysted 2024; Wu 2024; Firat 2026) are now used to set the Monte Carlo ranges of the FPV heat-loss coefficients, which include designs with no cooling advantage. PID studies on humidity and salt-induced surface conduction (Hoffmann 2014; Hacke 2015; Luo 2017; Islam 2018; Bora 2021; Li 2025) now support the leakage term.

Changes: Section I and Table I (new); Sections III-C, III-F, V-A; bibliography.

Section 16: Conclusions stronger than the evidence

Comment 13 (soften conclusions). *“Salinity is a stronger driver...” and “module selection is the most effective mitigation” should be presented as model-based findings.*

Response. Done. The conclusion now reads: “Under the baseline parameterization, salinity is the dominant driver of the predicted lifetime penalty,” and “Module selection for PID and salt-mist resistance is the most effective modelled mitigation, although its size rests on assumed reduction factors.” It closes by stating that the conclusions are model-based and call for field calibration.

Changes: Section VI; Section V-B.

Section 17: Figure 7

Comment 14 (model-derived). *Make the model-derived nature of the degradation components more prominent in Fig. 7.*

Response. The axis now reads “Modelled degradation rate”, and the caption states “Modelled module degradation rate by mechanism (baseline)” with a pointer to the assumed mitigation factors.

Section 18: Reproducibility

Comment 15 (reproducibility package). *Provide exact package versions, random seeds, configuration files, input weather, machine-readable parameters, unit tests and a single command that reproduces everything.*

Response. All are now provided:

- `requirements-lock.txt` pins the exact package versions (Python 3.11.15, TeX Live 2023).
- `paper/generated/parameters.json` records every scenario parameter, the Monte Carlo ranges and all seeds (baseline weather 42, ensemble 1–20, Monte Carlo 2024).
- The synthetic weather is regenerated deterministically from these seeds.
- 22 unit and regression tests cover, among other things, the reproducibility of the Monte Carlo and the agreement of the generator with its normals.
- `./reproduce.sh` regenerates every number, table and figure, the manuscript and this letter.

Sections 19–20: Recommended additional simulations

Comment 16 (Experiments A–D). *(A) Monte Carlo parameter uncertainty; (B) chloride threshold; (C) humidity threshold ΔT_d^* ; (D) mitigation comparison.*

Response. All four have been added:

- (A) 1,000 samples over 26 parameters, with medians, 95% intervals, probabilities of a positive difference and Spearman rank correlations (Table X, Fig. 12).
- (B) Saline degradation rises from 0.87%/yr at zero chloride to 1.31%/yr at 400 mg/m²/day. No chloride level in this range restores lifetime parity with land: even at zero chloride the deficit is -0.9%, because salt soiling, salt-assisted PID and the extra DC and availability losses remain.
- (C) Freshwater FPV loses its lifetime advantage above $\Delta T_d^* \approx 2.7$ K (Fig. 9b).
- (D) Table VIII compares six options for the saline plant: standard, PID-resistant, salt-mist-hardened, both, both plus glass-glass, and all plus 7-day cleaning. Each is reported with recovery, comparison with land and LCOE. Only with a lower-permeability package added does the saline plant exceed the land plant, under the stated assumptions.

Changes: Sections III-G, III-I, IV-E, IV-F, IV-H; Tables IV, VIII, X; Figs. 9 and 12.

Sections 21–22: Title and abstract

Comment 17 (title and positioning). *Shorten the title; position the abstract as a model-based assessment.*

Response. The reviewer’s second suggestion is adopted: “Lifetime Performance and Degradation of Floating Photovoltaic Systems Under Humid and Saline Conditions”. The abstract now opens

with “This study develops a coupled hourly simulation framework to quantify. . .”, reports the Monte Carlo intervals, and ends by stating that the results are model-based scenario predictions that require field validation.

Remaining limitation

The principal remaining weakness is the absence of measured validation data, both meteorological and FPV module-temperature, performance and degradation data. This revision quantifies how much the conclusions depend on the assumed parameters, but it cannot replace field data. The framework is ready to ingest measured hourly series, and calibration against monitoring data from operating FPV plants in coastal Andhra Pradesh is planned.